

Original Research

Multi-omics analysis unveils tumor heterogeneity and immunotherapy predictive model in breast cancer for precision medicine and early detection

Zhenxiong Zhao^{a,1}, Zhencang Zheng^{a,1}, Shenglu Jiang^a, Lingling Zhang^a, Xiufeng Tang^{b,*}

^a Taizhou Central Hospital (Taizhou University Hospital), Taizhou, Zhejiang 317000, China

^b Department of Pharmacy and Shandong Provincial key Traditional Chinese Medical Discipline of Clinical Chinese pharmacy, Shandong Cancer Hospital and Institute, Shandong First Medical University and Shandong Academy of Medical Sciences. Jinan, Shandong 255000, China

ARTICLE INFO

Keywords:

Breast cancer
Basal-like breast cancer
Tumor heterogeneity
Single-cell RNA sequencing
Immunotherapy
Prognostic model

ABSTRACT

Background: Intratumoral heterogeneity contributes to therapy resistance and immune evasion in breast cancer, making treatment strategies more complex. This study integrates single-cell RNA sequencing (scRNA-seq), spatial transcriptomics, and bulk RNA-seq deconvolution to characterize tumor subpopulations and develop a robust prognostic model.

Methods: We employed a multi-omics approach combining scRNA-seq, spatial transcriptomics, and bulk RNA-seq data deconvolution to explore the molecular diversity within breast cancer tumors. Tumor subtypes were identified based on distinct gene expression profiles, and functional pathway analysis was conducted to evaluate associations with clinical outcomes, including therapy resistance and immune evasion. Data from TCGA and GEO cohorts were integrated to validate the prognostic and immune-related findings. A CoxBoost+GBM algorithm was used to develop a robust prognostic model for patient survival and immunotherapy response prediction.

Results: Five distinct tumor subtypes were identified, each with unique functional profiles, underscoring the complexity of breast cancer heterogeneity. Basal-like breast cancer (BLBC) cells were found to play a central role in immune evasion and poor immunotherapy response, with high basal-like cell infiltration correlating with worse survival outcomes. Spatial transcriptomics revealed the widespread presence of BLBC cells across clinical subtypes, including ER+ tumors, suggesting their involvement in therapy resistance. A prognostic model based on CoxBoost+GBM demonstrated strong predictive power for patient survival and immunotherapy efficacy.

Conclusions: This study provides a comprehensive view of the genetic and immune determinants of breast cancer heterogeneity, with a focus on BLBC's role in immune escape and treatment resistance. These insights enhance the potential of multi-omics approaches in precision prevention, early detection, and personalized immunotherapy strategies.

Introduction

Breast cancer continues to be the most common cancer among women and a major contributor to cancer-related mortalities globally [1]. Despite advancements in early detection and multimodal treatment strategies, recurrence and metastasis undermine long-term survival [2]. While patients with nonmetastatic breast cancer achieve a 5-year overall survival (OS) rate exceeding 80%, this plummets to less than 25% for those with distant metastases [3]. Existing therapies, including surgery, radiotherapy, chemotherapy, targeted therapy, and endocrine therapy,

provide measurable benefits but are insufficient to address the persistent challenge of relapse, particularly in triple-negative breast cancer (TNBC) [4]. This underscores an urgent need for innovative therapeutic approaches to improve patient outcomes.

Immunotherapy has revolutionized the treatment landscape for multiple cancer types, with immune checkpoint inhibitors (ICIs) leading the charge [5]. An important development in the treatment of breast cancer was made in 2019 when the anti-programmed death-ligand 1 (PD-L1) antibody atezolizumab was approved for PD-L1-positive advanced TNBC [6]. However, clinical responses have remained

* Corresponding author.

E-mail address: tangxf927@126.com (X. Tang).

¹ These authors contributed equally to this work.

restricted to a subset of patients, with durability proving elusive [7]. For example, Pembrolizumab, an antibody that blocks PD-1, has an objective response rate (ORR) of 21.4% in TNBC; with subsequent treatment lines, this ORR drops to 5.7% [8]. There is a critical need for innovative methods to categorize patients likely to benefit from ICIs because existing biomarkers, such as PD-L1 expression and tumor-infiltrating lymphocytes, have limited predictive accuracy [9].

One of the major hurdles to immunotherapy efficacy in breast cancer is the inherent complexity of the tumor microenvironment (TME) [10]. The TME comprises a dynamic interplay of tumor cells, stromal components, immune cells, and extracellular matrix elements, collectively shaping tumor behavior and therapeutic response [11]. Tumor heterogeneity is a critical contributor to this complexity, encompassing phenotypic, epigenetic, transcriptional, metabolic, and microenvironmental diversity [12]. This heterogeneity drives immune evasion and therapeutic resistance, presenting a formidable barrier to durable immunotherapy responses [13]. While bulk RNA sequencing has provided valuable insights into the TME, its inability to resolve individual cellular subpopulations limits our understanding of tumor-immune dynamics.

Single-cell RNA sequencing (scRNA-seq) and spatial transcriptomics, the next-generation sequencing technologies, have revolutionized the ability to dissect the complexities of the TME at single-cell resolution [14]. These tools enable the identification of distinct tumor subpopulations and their spatial organization, offering unprecedented insights into the cellular and molecular mechanisms that underlie immune escape and therapy resistance.

In this study, we employed spatial transcriptomics and single-cell RNA sequencing (scRNA-seq) to understand the complexity of breast cancer heterogeneity. Our findings highlight the BLBC as a key driver of immune escape and poor prognosis. We developed a Prognostic Index Score (PIS) model based on basal-like markers, effectively predicting patient risk and immunotherapy response. These insights provide a foundation for improving breast cancer stratification and tailoring immunotherapy strategies.

Materials and methods

Acquisition and processing of transcriptome data

RNA expression profiles and clinical data ($n=1097$) from the TCGA-BRCA dataset were obtained and split equally into training and validation cohorts. The training cohort was used to develop the prognostic model, while the validation cohort assessed its stability and accuracy. Data were normalized to TPM format, log2-transformed, and subjected to batch effect correction with the SVA package's Combat algorithm. To ensure the robustness of our prognostic model, the baseline clinical characteristics between the training and validation cohorts were compared. No statistically significant differences were observed in key clinicopathological parameters, including age, tumor stage, tumor grade, and PAM50 molecular subtypes. Additionally, two untreated RNA-seq datasets from GEO (GSE22093 and GSE22513) served as external validation sets.

Acquisition and processing of scRNA-seq data

Single-cell RNA sequencing data were retrieved from the GEO dataset GSE176078, encompassing 26 untreated primary tumor samples (10 triple-negative breast cancers, 5 HER2-positive, and 11 ER-positive breast cancers). Data was processed using R (v4.1.3) and Scanpy (v1.9.1). The following criteria were used to filter the cells: UMI counts between 200 and 15000, detected gene counts between 200 and 4500, mitochondrial gene content $< 12\%$, and blood cell content $< 5\%$. BBKNN was used to correct batch effects after 3000 highly variable genes were identified and normalized, and UMAP was used to reduce dimensionality. Seurat (v5.0) with the FindAllMarkers function was

used to identify differentially expressed genes (DEGs) based on the following thresholds: $p < 0.05$, $\log_2FC < 1$, and $> 30\%$ expression frequency. Downstream analysis, including Harmony-based batch correction and SCTransform normalization, was performed.

Cell annotation

Cell types were annotated using canonical marker genes. Epithelial cells were identified by EPCAM, KRT18, KRT19, and CDH1; endothelial cells by PECAM1, CLDN5, FLT1, and RAMP2; fibroblasts by DCN, THY1, COL1A1, and COL1A2; and immune subsets including T cells (CD3D, CD3E, CD3G, TRAC), B cells (CD79A, IGHM, IGHG3, IGHA2), NK cells (NKG7, GNLY, NCAM1, KLRD1), and myeloid cells (LYZ, MARCO, CD68, FCGR3A). Mast cells were characterized by KIT, MS4A2, and GATA2. Following this initial broad classification, a critical step was taken to ensure the purity of the tumor cell population for downstream analysis: all non-epithelial cells (endothelial, fibroblast, and immune cells) were excluded. The analysis of tumor heterogeneity and the identification of neoplastic subtypes (Basal, HER2+, LumA, LumB, Cycling) were performed exclusively on this purified epithelial cell population. Annotations for epithelial subpopulations were refined using publicly available data from CellxGene (<https://cellxgene.cziscience.com>), ensuring detailed classification.

Prognostic gene identification

Prognostic genes were identified via univariate Cox regression analysis using TCGA data, yielding 50 genes with $p < 0.05$. These were subsequently validated in internal and external cohorts to develop a Prognostic Index Score (PIS) model.

Development of tumor-associated risk features

Risk scores were computed using 101 machine learning algorithms, with the optimal cutoff values determined using the `surv_cutpoint` function. Patients were classified into high- and low-risk groups, and the model's predictive accuracy was evaluated using survival metrics.

Machine learning-enhanced prognostic model construction

A prognostic model was developed through a comprehensive machine-learning workflow designed to maximize robustness and minimize overfitting. The entire TCGA-BRCA cohort was first randomly partitioned into a training cohort ($n=548$) for model construction and a held-out internal validation cohort ($n=549$) which was strictly reserved for the final, one-time assessment of the selected model. To prevent overfitting during the model selection process, we employed a rigorous nested validation strategy where all model development and algorithm comparison were confined to the training set. Specifically, we evaluated 101 algorithmic combinations derived from ten machine learning algorithms, including Random Survival Forest (RSF), Elastic Net (Enet), Lasso, Ridge, stepwise Cox, CoxBoost, Cox Partial Least Squares Regression (plsRcox), Supervised Principal Components (SuperPC), Generalized Boosted Regression Models (GBM), and Survival Support Vector Machines (survival-SVM). The performance of each algorithm combination was evaluated and its hyperparameters were tuned using Leave-One-Out Cross-Validation (LOOCV) exclusively on the training cohort. This LOOCV process provided a nearly unbiased estimate of the model's generalizability for each candidate, and the combination achieving the highest average C-index in this internal cross-validation was selected as the final model. The finalized model was then frozen and applied to the independent held-out validation cohort and external GEO datasets to generate all reported performance metrics, thereby ensuring that the results reflect true predictive power rather than overfitting to the training data.

Pseudotime analysis

Pseudotime trajectories were constructed for tumor subpopulations using Monocle2 with DDRTree for dimensionality reduction. CytoTRACE was employed to assess differentiation states.

Mutation and CNV analysis

Copy number variations (CNVs) were analyzed using GISTIC 2.0, which identifies recurrent regions of amplification or deletion across the genome. Tumor mutational burden (TMB) was calculated using maf-tools, a package designed to analyze mutation data and assess the total number of somatic mutations in tumor samples per megabase. CNV and TMB profiles were then correlated with the Prognostic Index Score (PIS) to evaluate their association with patient outcomes and tumor aggressiveness.

Immune correlation analysis

Immune infiltration was quantified using IOBR, incorporating seven computational methods. The estimate R package was used to determine stromal, immune, and tumor cell proportions, which were compared across risk groups.

Bulk data deconvolution

Tumor subpopulation proportions in bulk RNA-seq datasets were inferred using the BisqueRNA package with scRNA-seq-derived markers. Subtype-specific scores were computed using GSVA.

Prediction of immunotherapy response

Immune therapy responses were predicted using the TIDE algorithm in the TCGA and GEO datasets. To evaluate differences in immunotherapy efficacy, PIS scores were assessed across multiple datasets (e.g., GSE100797, GSE115821, GSE145996, GSE126044).

Spatial transcriptomics data analysis

Spatial transcriptomics data were obtained from the public repository Zenodo (accession: <https://zenodo.org/records/4739739>). From this dataset, four samples—specifically CID44971 and CID4465 (triple-negative breast cancer, TNBC), along with CID4535 and CID4290 (ER-positive breast cancer)—were selected for subsequent analysis. To accurately resolve the cellular composition within each Visium capture spot and enable the distinction between tumor-infiltrating lymphocytes and adjacent stromal cells, we performed an integrative analysis with a well-annotated single-cell RNA sequencing reference. The distinct transcriptional signatures of major cell types—including epithelial cells, T cells, B cells, myeloid cells, fibroblasts, and endothelial cells—derived from the scRNA-seq data were leveraged to deconvolute the spot-level spatial transcriptomic profiles using a robust computational approach.

Clinical tissue collection and processing

Primary tumor tissues and matched adjacent normal tissues (located >3 cm from the tumor margin) were obtained from five treatment-naïve breast cancer patients who underwent curative resection at Taizhou University Hospital. Immediately after surgical excision, specimens were snap-frozen in liquid nitrogen and stored at -80°C until RNA extraction. The study was approved by the Institutional Review Board of Taizhou University Hospital, and written informed consent was obtained from all participants prior to tissue collection.

RNA extraction and quantitative real-time PCR (qRT-PCR)

Total RNA was extracted using TRIzol reagent (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's protocol. RNA purity and concentration were determined using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific), with acceptable A260/A280 ratios ranging from 1.8 to 2.0. First-strand cDNA was synthesized from 1 μg of total RNA using the PrimeScript RT Reagent Kit (Takara, Japan). Quantitative real-time PCR (qRT-PCR) was performed using SYBR Premix Ex Taq (Takara) on a QuantStudio 5 Real-Time PCR System (Applied Biosystems, USA). The thermal cycling conditions were as follows: initial denaturation at 95°C for 30 seconds, followed by 40 cycles of 95°C for 5 seconds and 60°C for 30 seconds. Gene expression levels were normalized to GAPDH and calculated using the $2^{-\Delta\Delta\text{Ct}}$ method. All reactions were performed in triplicate.

Cell culture and authentication

All cell lines were authenticated by short tandem repeat (STR) profiling and routinely screened for mycoplasma contamination. Cells were cultured in Dulbecco's Modified Eagle Medium (DMEM; Gibco, Grand Island, NY, USA) supplemented with 10% fetal bovine serum (FBS; Gibco) and 1% penicillin-streptomycin (Gibco), and maintained at 37°C in a humidified atmosphere with 5% CO_2 . For PIS expression profiling, cells were harvested at 80% confluence and subjected to qRT-PCR as described above. Experiments were conducted in triplicate with independent biological replicates.

siRNA transfection

Small interfering RNAs (siRNAs) targeting PIS and a negative control siRNA were dissolved in nuclease-free water to a final concentration of 10 μM . MCF7 and MDA-MB-436 cells were seeded in 6-well plates at a density of 2×10^5 cells/well and incubated overnight. Transfection complexes were prepared by mixing 50 nM siRNA with 5 μL of Lipofectamine 3000 (Invitrogen, L3000015) in Opti-MEM medium (Gibco, 31985070), following the manufacturer's protocol. After 15 minutes of incubation at room temperature, the complexes were added to the cells. The medium was replaced with fresh complete culture medium 6 hours after transfection. At 48 hours post-transfection, total RNA was extracted for cDNA synthesis and qRT-PCR to assess knockdown efficiency, which was quantified using the $2^{-\Delta\Delta\text{Ct}}$ method. Efficient knockdown (>70%) was confirmed relative to negative control ($p < 0.01$, Student's t-test). All experiments were repeated independently in triplicate.

Cell proliferation assay (CCK-8)

For cell proliferation analysis, 3×10^3 transfected cells were seeded per well in 96-well plates. At 24, 48, 72, and 96 hours post-transfection, 10 μL of Cell Counting Kit-8 (CCK-8) reagent (Dojindo, Japan) was added to each well, followed by 2 hours of incubation at 37°C . Absorbance was measured at 450 nm using a microplate reader (BioTek Synergy H1). Each group contained five technical replicates, and proliferation curves were generated by normalizing absorbance values to the 0-hour time point.

Apoptosis assay by flow cytometry

Apoptosis was evaluated 48 hours post-transfection using Annexin V-FITC/propidium iodide (PI) staining. Cells were harvested, washed twice with PBS, and stained using the Annexin V Apoptosis Detection Kit (BD Biosciences, USA) per the manufacturer's instructions. Stained cells were analyzed immediately on a BD FACSVerser flow cytometer. Early and late apoptotic populations were defined as Annexin V+/PI− and Annexin V+/PI+, respectively. Flow cytometry data were processed using FlowJo v10 software.

Migration and invasion assays

Cell migration and invasion capabilities were assessed using Transwell chambers (8-μm pore size; Corning, USA). For migration assays, 5 × 10⁴ cells in serum-free medium were seeded into the upper chambers, while the lower chambers were filled with medium containing 10% FBS as a chemoattractant. After 24 hours, non-migrated cells on the upper surface were removed, and migrated cells on the underside of the

membrane were fixed with 4% paraformaldehyde, stained with 0.1% crystal violet, and counted in five randomly selected microscopic fields per well. For invasion assays, the upper chambers were pre-coated with Matrigel (1:8 dilution in DMEM; BD Biosciences) and incubated at 37°C for 4 hours before cell seeding. The remainder of the protocol was identical to the migration assay.

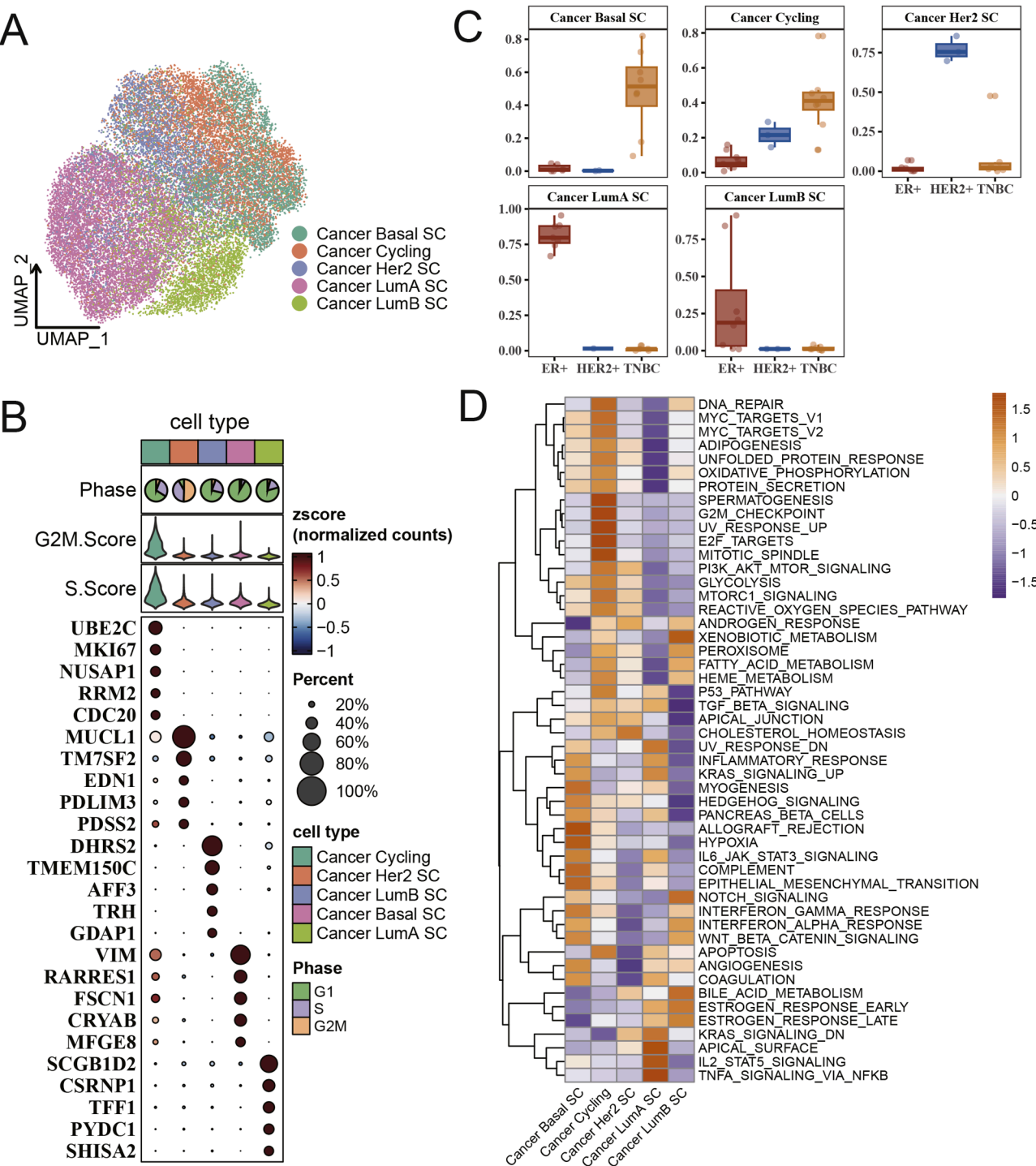


Fig. 1. Identification and characterization of neoplastic subtypes in breast cancer. (A) UMAP plot showing five major neoplastic cell subtypes identified via scRNA-seq, including Basal, HER2+, LumA, LumB, and Cycling subtypes. (B) Bubble plot of marker gene expression specific to each neoplastic subtype. (C) Box plots depicting the relative abundance of neoplastic subtypes across HER2+, ER+, and TNBC samples. (D) Heatmap displaying ssGSEA scores for 50 hallmark pathways across the five subtypes.

Western blotting

Total cellular protein was extracted using RIPA lysis buffer (Beyotime, China) supplemented with protease inhibitors (Roche). Protein concentration was determined using a BCA Protein Assay Kit (Pierce). Equal amounts of protein (30 μ g per sample) were separated by 10% SDS-PAGE and transferred to PVDF membranes (Millipore). After blocking with 5% non-fat milk for 1 hour at room temperature, membranes were incubated overnight at 4°C with the following primary antibodies: cleaved Caspase-3 (1:1000, #9664, CST), E-Cadherin

(1:2000, #3195, CST), Bcl-2 (1:1000, #15071, CST), Vimentin (1:1000, #5741, CST), and β -actin (1:5000, #4970, CST). HRP-conjugated secondary antibodies (1:5000, CST) were applied for 1 hour at room temperature. Protein bands were visualized using enhanced chemiluminescence (ECL; Millipore) and quantified using ImageJ software.

Statistical analysis

R (v4.1.3) was used for statistical analysis and visualizations.

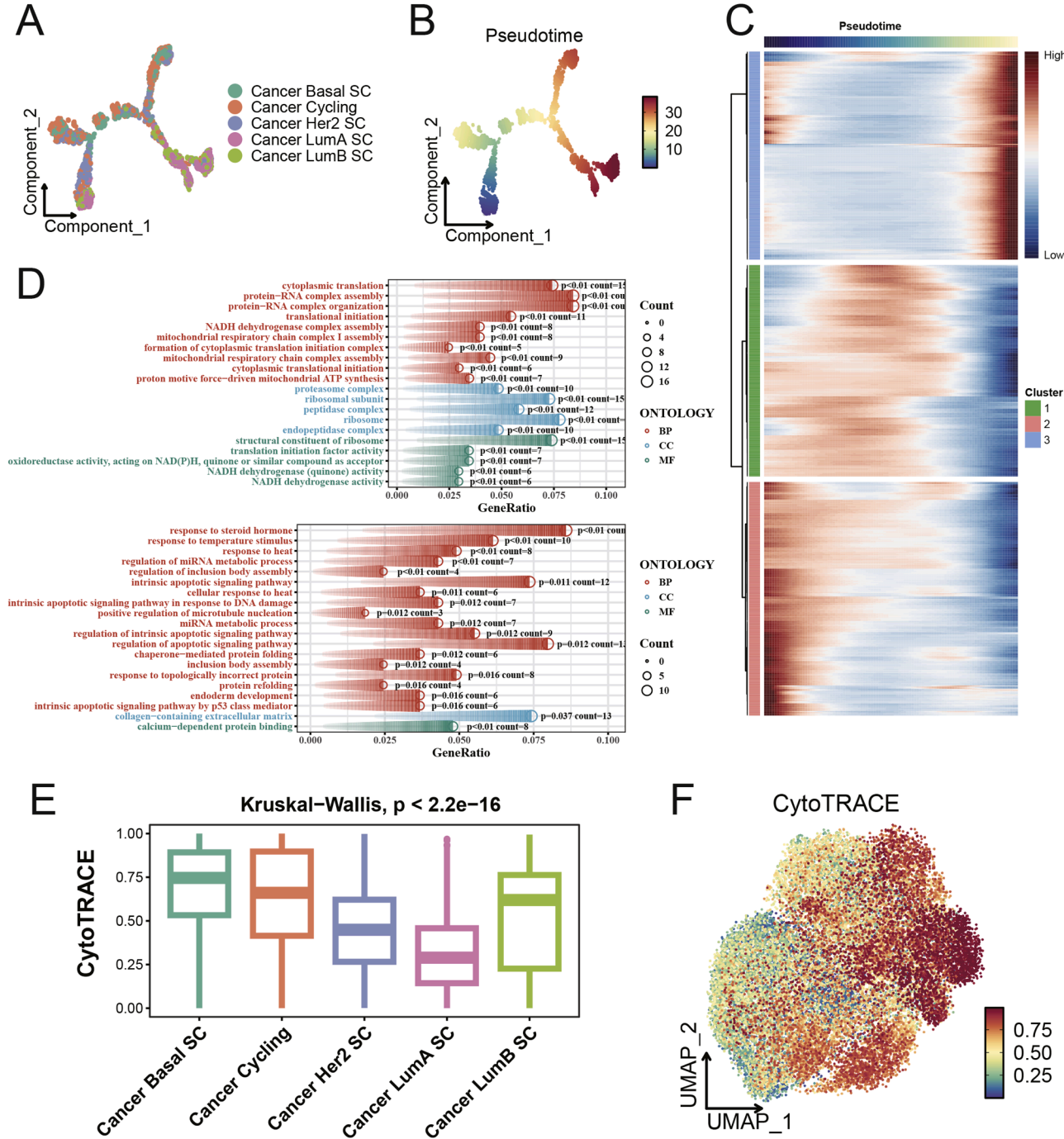


Fig. 2. Pseudo-time trajectory analysis and differentiation potential of neoplastic subtypes. (A-B) Analysis of pseudo-time trajectories based on monoclones illustrating the differentiation states among neoplastic subtypes. (C) Heatmap of pseudo-time-associated marker genes clustered into three groups (C1–C3), with C2 representing early differentiation and C3 terminal states. (D) GO enrichment analysis identifying biological processes associated with early (C2) and late (C3) pseudo-time stages. (E-F) CytoTRACE analysis showing differences in differentiation potential across neoplastic subtypes.

Categorical data were compared using Chi-square testing, while continuous variables were assessed using Pearson correlation. The Wilcoxon rank-sum test or T-test was used for continuous variables. Optimal cutoffs were determined using the survminer package, and Kaplan-Meier survival curves were analyzed with the survival package.

Results

ScRNA-seq reveals intratumoral heterogeneity in breast cancer

To investigate intrinsic tumor heterogeneity, we first performed cell

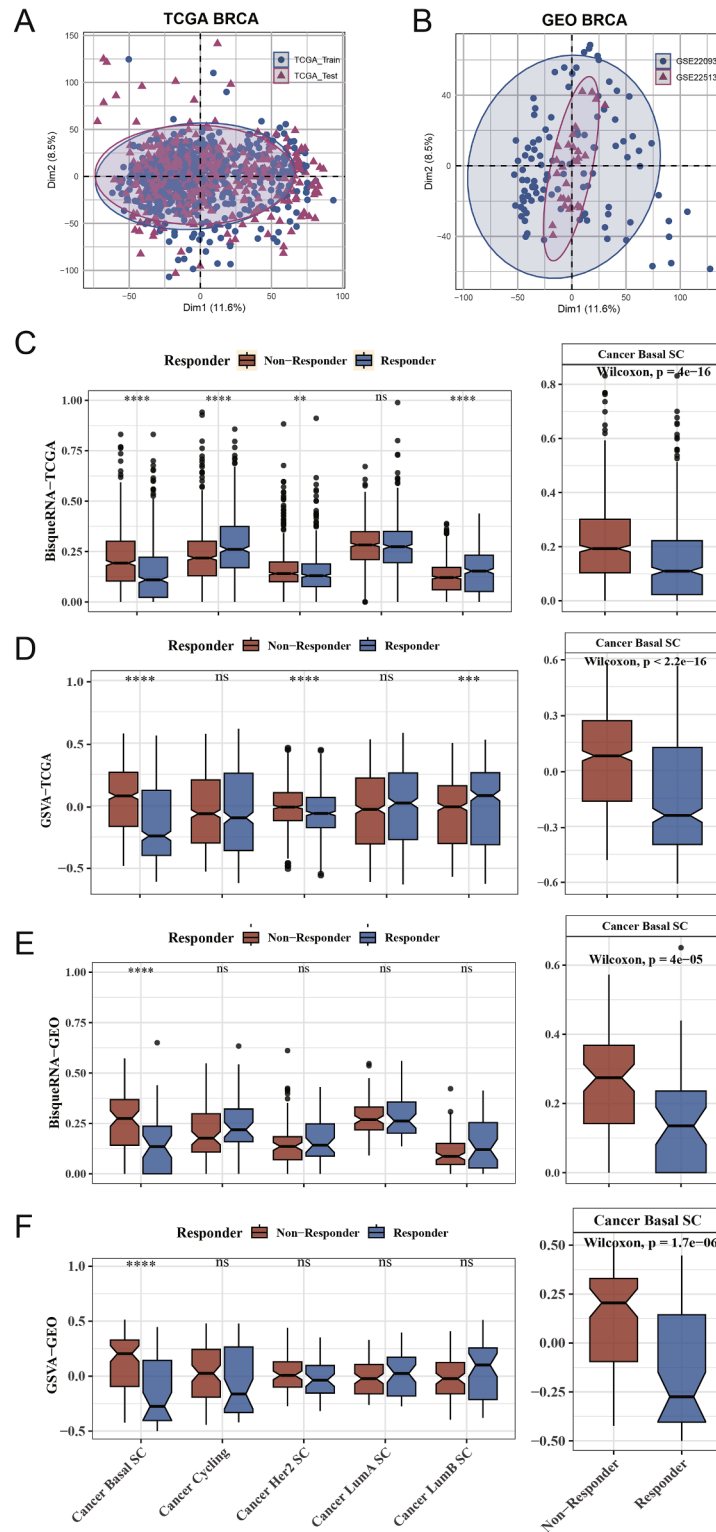


Fig. 3. Immune response stratification and deconvolution of bulk RNA-seq datasets using single-cell-defined subtype markers. (A-B) PCA plots confirming successful batch effect correction in TCGA and GEO datasets using the Combat algorithm. (C and E) Box plots comparing subtype composition scores, estimated via BisqueRNA, between immunotherapy responder and non-responder groups in bulk RNA-seq and microarray datasets. (D and F) Box plots of GSVA scores, derived from subtype marker genes, showing significant differences between immunotherapy responder and non-responder groups in bulk RNA-seq and microarray datasets.

type annotation on the scRNA-seq data, which identified major lineages including epithelial, immune, and stromal cells. We then focused our subsequent analysis specifically on the epithelial cell compartment, which encompasses the neoplastic cells, to avoid confounding signals from the tumor microenvironment. Graph-based clustering identified five distinct tumor subtypes based on canonical markers: Basal, HER2+, LumA, LumB, and Cycling subtypes (Figs 1A-B). Notably, these molecular subtypes demonstrated heterogeneity not fully captured by conventional clinical classifications (HER2+, ER+, TNBC), as Cycling cells were distributed across all clinical subtypes (Fig. 1C). Functional annotation via ssGSEA further revealed unique pathway activity among the subtypes. For example, the Basal subtype showed notable enrichment in immune-related pathways such as Inflammatory Response, Complement, and Interferon Gamma Response, which differed

significantly from other tumor subtypes (Fig. 1D).

Developmental trajectories highlight functional hierarchies in tumor subpopulations

Building upon the identified subtypes, we explored their developmental relationships using Pseudotime trajectory analysis with Monocle2. The results revealed three major developmental branches, with Basal and Cycling cells sharing a common trajectory, suggesting developmental or functional links (Figs 2A-B). Gene expression dynamics along pseudotime showed distinct early- and late-stage gene clusters (Fig. 2C). Early-stage genes were enriched in translational initiation and ribosomal subunit assembly, while late-stage genes were associated with apoptotic signaling and extracellular matrix

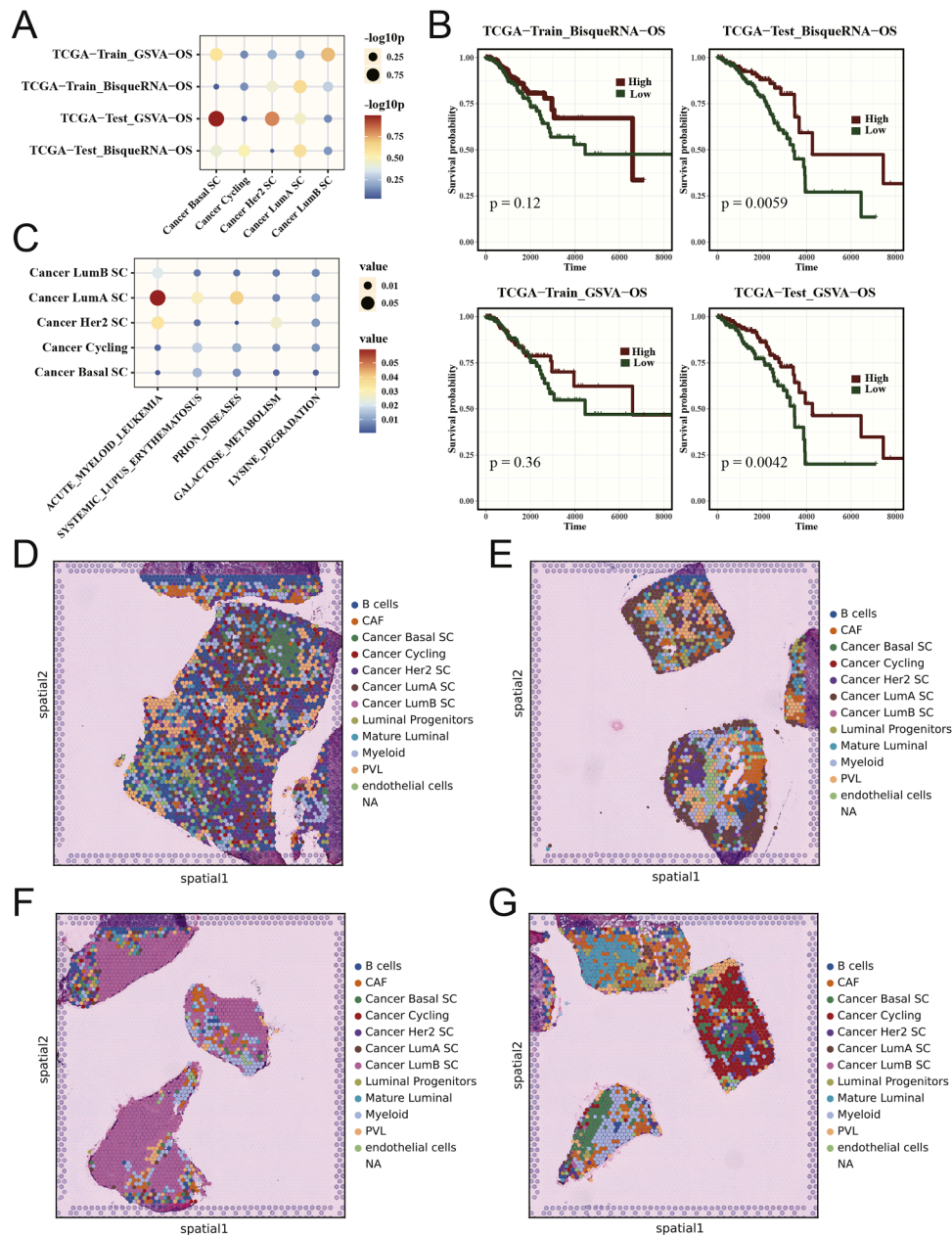
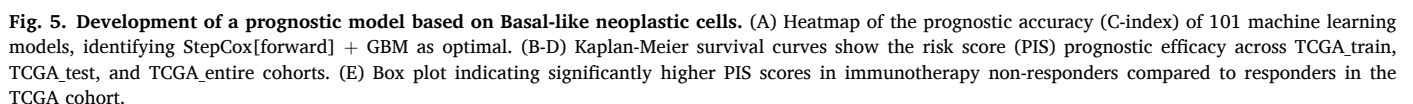


Fig. 4. Prognostic relevance and spatial localization of neoplastic subtypes. (A) Bubble plot correlating subtype composition and neoplastic marker genes related GSVA scores with overall survival (OS) in TCGA_train and TCGA_test cohorts. (B) Kaplan-Meier survival curves for Basal-like neoplastic cell composition and GSVA scores across TCGA_train and TCGA_test cohorts. (C) Bubble plot showing metabolic pathway enrichment scores for each neoplastic subtype. (D-G) Spatial transcriptomics data from four breast cancer samples display stromal, immune, and neoplastic subtype distributions; (D, F) represent ER+ samples, and (E, G) represent TNBC samples.

To better understand the clinical significance of tumor subpopulations, we examined their immune characteristics in TCGA-BRCA and two GEO cohorts (GSE22093 and GSE22513). Batch effect



correction with Combat ensured consistency across datasets (Figs 3A-B). Using BisqueRNA and GSVA, we calculated subtype proportions and assessed their relationship with immune escape scores via the TIDE algorithm. Among the subtypes, basal-like cells demonstrated the strongest association with immune escape and poorer predicted immunotherapy responses (Figs 3C-F). Building on these findings, survival analyses revealed that patients with higher proportions of Basal-like cells had significantly worse overall survival (Figs 4A-B), emphasizing their potential as a prognostic marker. To further investigate the spatial context of these subtypes, spatial transcriptomics analysis uncovered substantial intratumoral heterogeneity. Notably, Basal-like cells were detected even in ER+ tumors, indicating their widespread presence

across clinical subtypes and potential role in driving aggressive tumor behavior (Figs 4D-G).

Development and validation of a prognostic model related to basal-like neoplastic cells

Recognizing the strong prognostic relevance of Basal-like neoplastic cells, we developed a Prognostic Index Score (PIS) model using marker genes from this subtype. Among 101 tested algorithm combinations, the Stepwise Cox (forward) + GBM algorithm achieved the highest C-index (Fig. 5A). The PIS robustly stratified patients into risk groups in TCGA training, validation, and full cohorts, with higher scores consistently

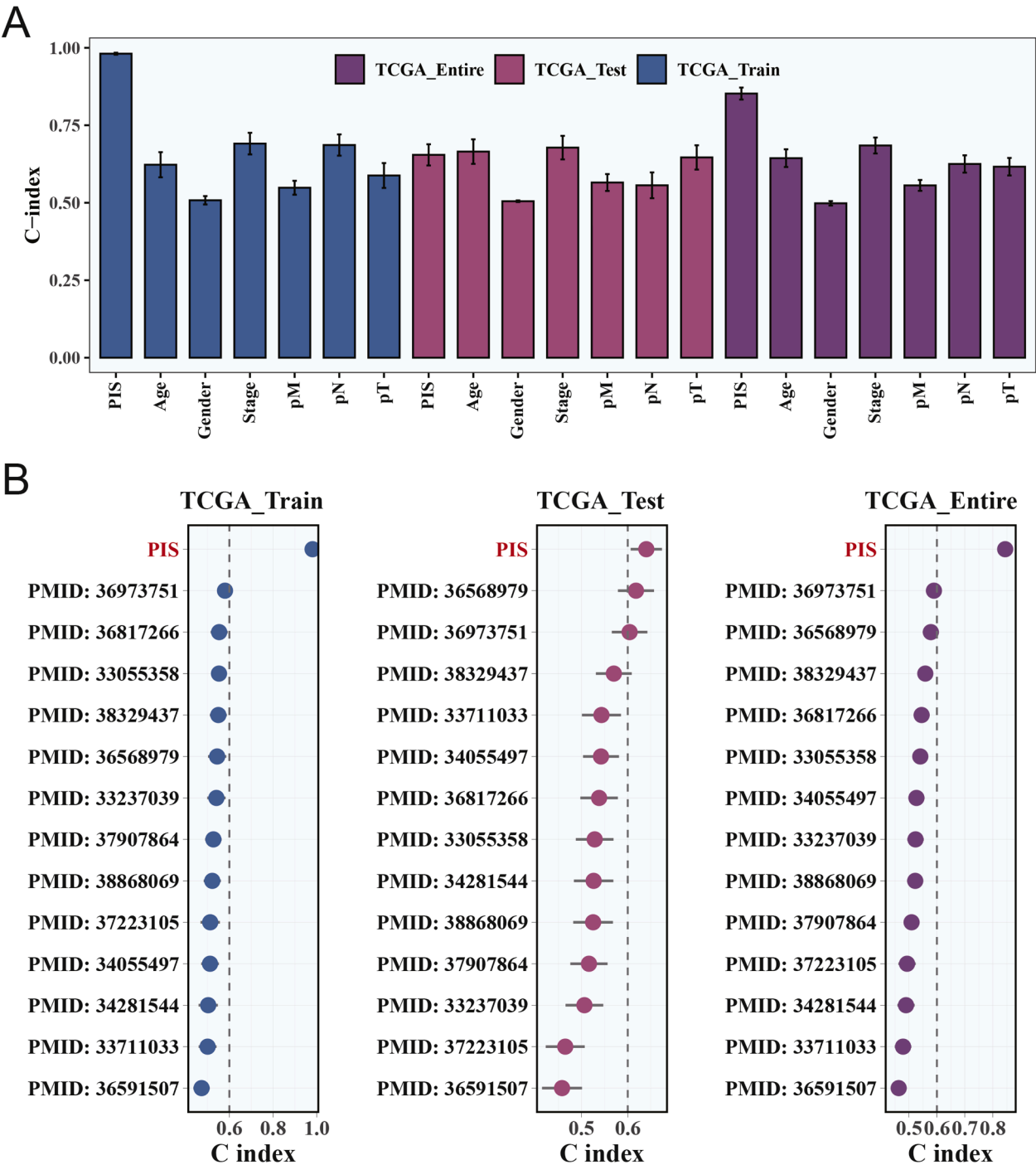


Fig. 6. Comparison of the prognostic model with clinical indicators and published models. (A) Bar chart comparing PIS model C-index values with traditional clinical indicators across cohorts. (B) C-index plot comparing the PIS model with previously published prognostic models.

correlating with poorer OS (Figs 5B-D). Notably, PIS was higher in immune-nonresponsive groups, indicating its potential to capture immune-tumor dynamics (Fig. 5E). Moreover, PIS outperformed traditional clinical factors and existing prognostic models in stratifying patient risk, demonstrating superior predictive power (Figs 6A-B).

Genomic variations and tumor mutational burden about PIS

Tumor mutational burden (TMB) reflects the generation of neo-antigens, serving as a predictive marker for immunotherapy efficacy across various cancers [15]. This study investigated the relationship

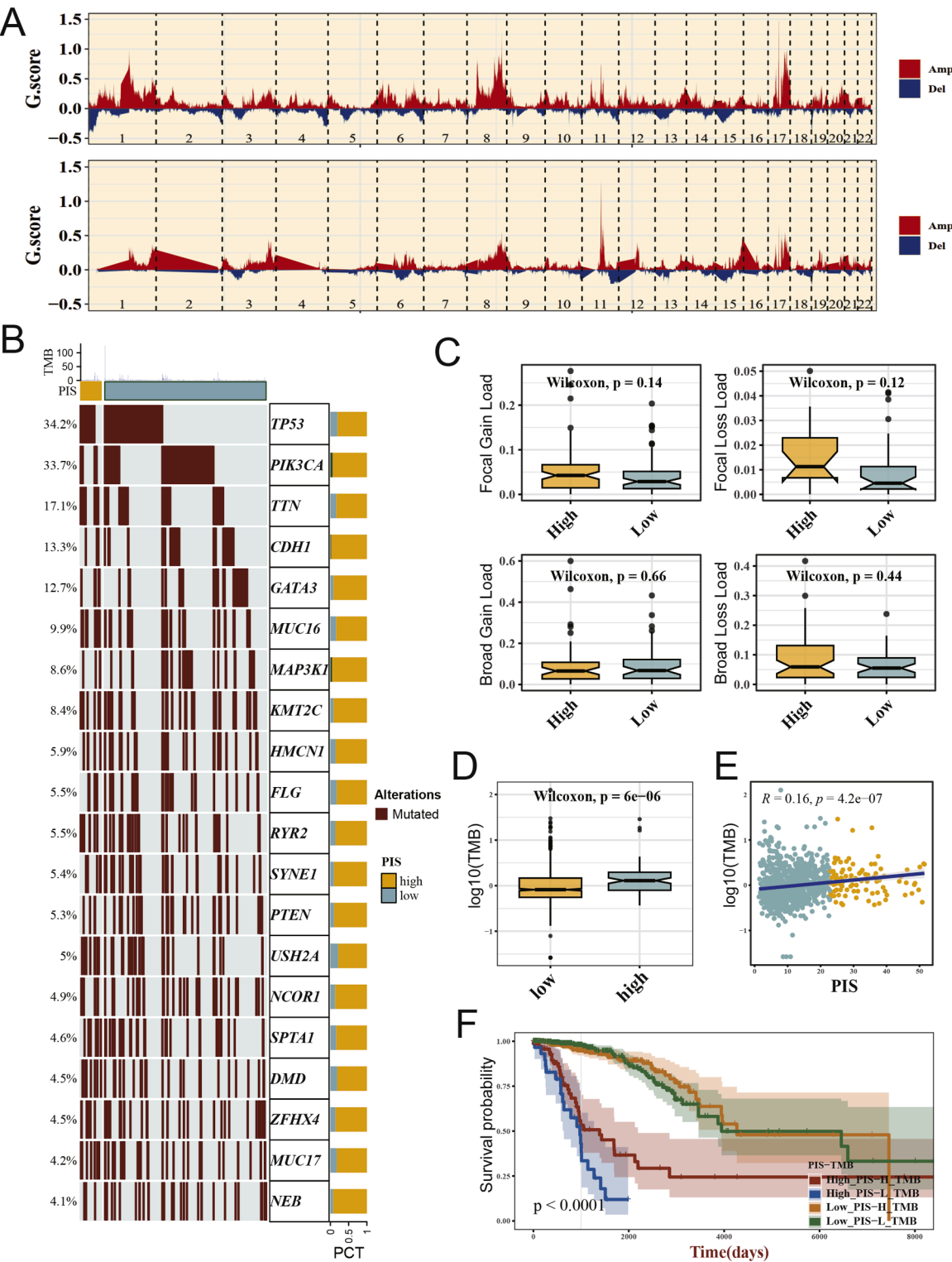


Fig. 7. Genomic variations and tumor mutational burden (TMB) associated with PIS. (A) Box plot comparing amplification (AMP) and deletion (DEL) events between high- and low-PIS groups. (B) Waterfall plot displaying CNV differences between high- and low-PIS groups. (C) Box plots showing differences in focal and broad genomic alterations between PIS groups. (D) Box plot showing TMB differences and correlation between TMB (log10 scale) and PIS. (E) Kaplan-Meier survival curves for combined PIS and TMB scores.

between PIS and genomic alterations, including copy number variations (CNVs) and TMB. A comparative analysis of CNVs between high- and low-PIS groups revealed significant amplifications among high-PIS patients in the 1q, 8q, and 17q chromosomal regions (Fig. 7A). Additionally, mutation profiling demonstrated that both PIS groups shared high frequencies of mutations in TP53 (34.2%), PIK3CA (33.7%), and TTN (17.1%), emphasizing the critical role of these genes in driving breast cancer pathogenesis and influencing patient prognosis (Fig. 7B). Notably, while no significant differences in the patterns of focal or broad

chromosomal gains and losses were observed between PIS groups (Fig. 7C), a strong positive correlation emerged between PIS and TMB when analyzed on a log10 scale (Figs 7D-E). Patients with high PIS scores and elevated TMB exhibited the poorest overall survival, as shown by Kaplan-Meier analysis (Fig. 7F). These findings suggest that high PIS tumors are characterized by increased genomic instability, potentially contributing to their aggressive clinical behavior and diminished responsiveness to conventional and immune-targeted therapies.

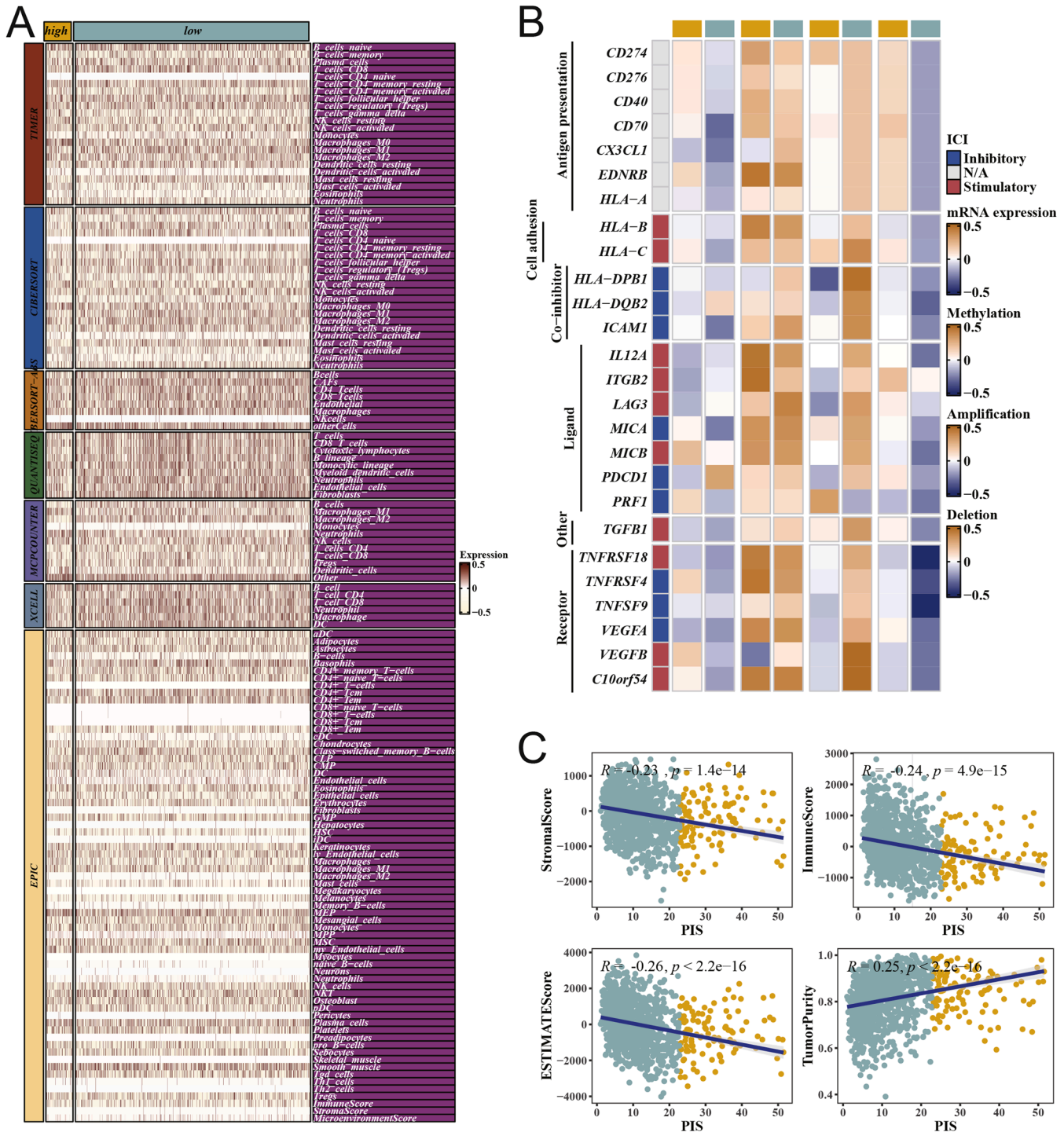


Fig. 8. Immune cell infiltration and immune-related gene alterations associated with PIS. (A) Heatmap of immune cell infiltration differences across high- and low-PIS groups, calculated using seven algorithms. (B) Heatmap showing expression, CNV alterations, and methylation of immune-related genes correlated with PIS. (C) Correlation analysis showing a positive relationship between PIS and tumor purity (ESTIMATE-derived scores).

To further characterize the immune microenvironment of high- and low-PIS tumors, we analyzed immune cell infiltration using seven algorithms. Both groups demonstrated low infiltration of naïve T cells and

NK cells (Fig. 8A). Further analysis of immune-related gene expression, methylation, and copy number variations (CNVs) revealed substantial immune activation in the high PIS group. Specifically, this group showed higher expression of immune checkpoints (CD274, CD276...) (Fig. 8B). Correlation analysis with ESTIMATE scores further reinforced these

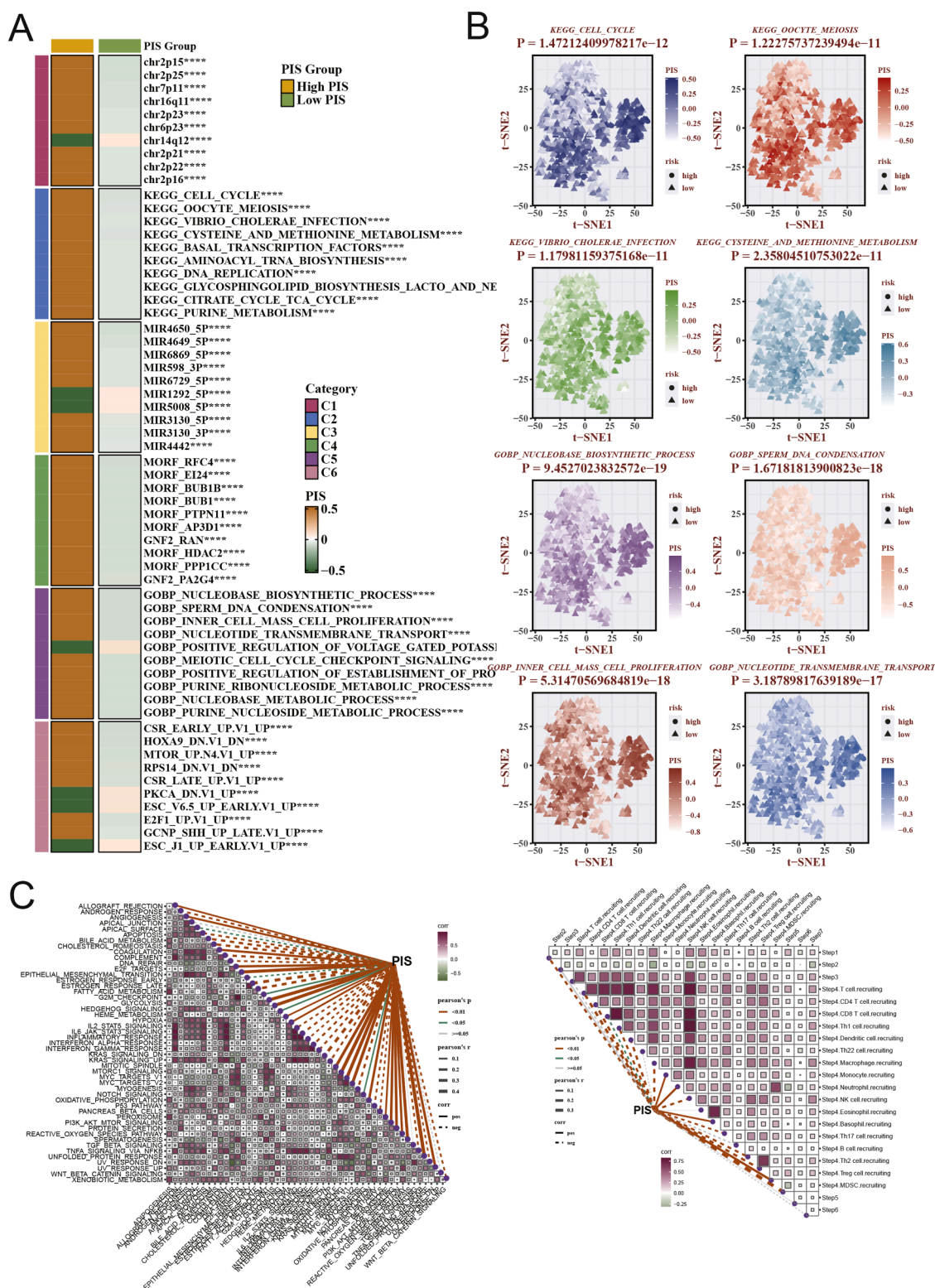


Fig. 9. Functional pathway enrichment analysis in high- and low-PIS groups. (A) Heatmap of functional signatures (Msigdb C1–C6 categories) with ssGSEA scores comparing groups with high and low PIS. (B) t-SNE plots visualizing key differences in GOBP and KEGG pathways between PIS groups. (C) Bubble plot illustrating correlations of PIS with hallmark pathways and immune-related pathways (TIP dataset).

observations, showing negative correlations between PIS and Stromal-Score, ImmuneScore, and ESTIMATEScore, alongside a positive correlation with TumorPurity (Fig. 8C). To elucidate the functional implications of PIS, we performed pathway enrichment analysis using ssGSEA. According to pathway enrichment analysis, high-PIS tumors were more prevalent in pathways linked to proliferation, including nucleobase biosynthesis and the cell cycle (Fig. 9A-B). Additionally, A broader analysis of hallmark pathways and tumor immune infiltration (via TIP) revealed that high-PIS tumors had greater Treg infiltration and

fewer antitumor immune cells (Fig. 9C).

Correlation analysis with immune-related genes and implications for immunotherapy

We analyzed PIS' correlation with immune-related genes to assess its utility in predicting the efficacy of immunotherapy. A bubble plot revealed strong positive correlations between PIS and multiple immune-related genes (Fig. 10A). Further analysis of key immunotherapy targets,

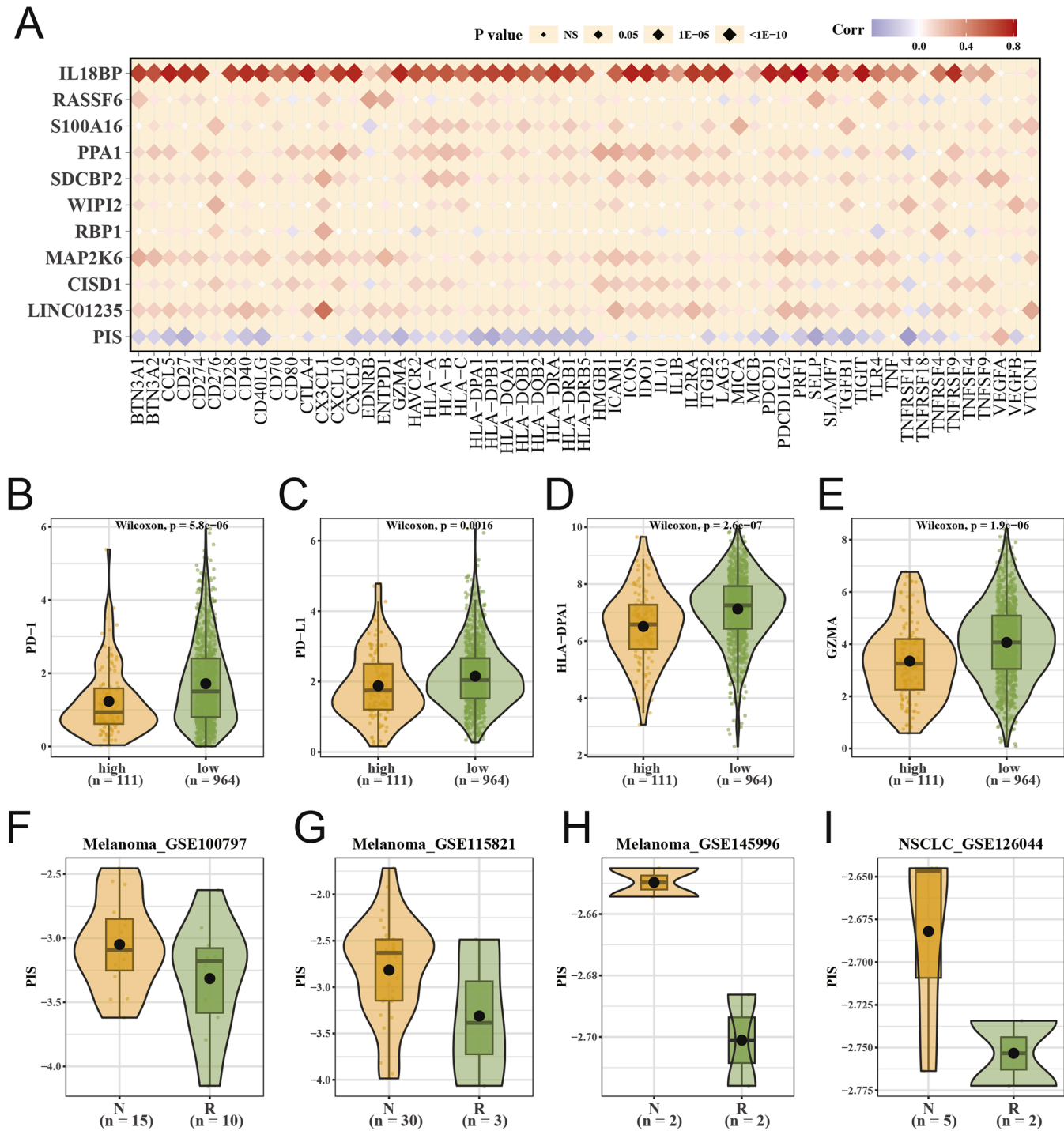


Fig. 10. Immune response analysis and implications for immunotherapy based on PIS. (A) Bubble plot showing correlations between PIS-associated genes and immune-related genes. (B-E) Violin plots comparing expression of immune checkpoint genes (e.g., PDCD1, CD274) between high- and low-risk PIS groups. (F-I) Violin plots indicate significantly higher PIS scores in immunotherapy non-responders across multiple cohorts, supporting its predictive value for immunotherapy response.

including PDCD1 (PD-1), CD274 (PD-L1), HLA-DPA1, and GZMA, demonstrated significantly higher expression levels of these genes in the low-PIS group compared to the high-PIS group (Figs 10B–E). This differential expression suggests tumors with lower PIS may exhibit a more favorable immune microenvironment conducive to immunotherapy responses. To directly assess the predictive value of PIS for immunotherapy efficacy, we analyzed PIS scores across four independent immunotherapy cohorts (GSE100797, GSE115821, GSE145996, and GSE126044). In every cohort, non-responders consistently showed noticeably higher PIS scores than responders (Figs 10F–I). These findings strongly indicate that patients with lower PIS scores are more likely to benefit from immune checkpoint inhibitors and other immunotherapy modalities.

PIS promotes breast cancer progression by enhancing proliferation, survival, and invasive potential

To experimentally validate the oncogenic role of PIS, a key gene identified in our prognostic model, we assessed its expression pattern and biological functions in breast cancer. Transcriptomic analysis revealed significantly elevated PIS mRNA levels in breast tumor tissues compared to matched adjacent normal tissues (Fig. 11A), implicating its potential involvement in tumorigenesis. We further examined PIS expression across a panel of breast cancer cell lines, including MDA-MB-231, T-47D, MCF7, MDA-MB-436, and BT-483. Among these, MCF7 and MDA-MB-436 exhibited the highest expression levels and were thus selected for subsequent functional assays (Fig. 11B). To investigate the biological role of PIS, we performed siRNA-mediated knockdown in both MCF7 and MDA-MB-436 cells. qRT-PCR confirmed efficient silencing of PIS expression (Fig. 11C). Cell proliferation assays using CCK-8 over a four-day period demonstrated a significant reduction in proliferation following PIS knockdown, indicating its role in promoting breast cancer cell growth (Figs 11D–E). Flow cytometric analysis of apoptosis revealed a marked increase in apoptotic cell populations in MDA-MB-436 cells upon PIS silencing, highlighting its pro-survival function (Figs 11F–G). Furthermore, transwell migration and invasion assays showed that PIS knockdown significantly impaired both migratory and invasive capacities of MDA-MB-436 cells, supporting its role in enhancing tumor aggressiveness (Figs 11H–I). Western blot analysis corroborated these findings at the protein level, showing increased expression of cleaved caspase-3 and E-cadherin—markers of apoptosis and epithelial phenotype, respectively—and decreased expression of Bcl-2 and Vimentin, which are associated with cell survival and mesenchymal traits (Figs 11J–K).

Discussion

This study comprehensively explores intratumoral heterogeneity in breast cancer using scRNA-seq and spatial transcriptomics. By examining distinct tumor subpopulations, particularly BLBC cells, we underscore their significant prognostic and therapeutic implications, especially within immunotherapy.

Tumor heterogeneity remains a major challenge in breast cancer, driving immune evasion, therapy resistance, and disease progression [16–18]. Our scRNA-seq analysis identified five tumor subtypes—Basal, HER2+, LumA, LumB, and Cycling—each characterized by distinct functional and pathway activity profiles. Importantly, these molecular subtypes extend beyond conventional clinical classifications (ER+, HER2+, TNBC), providing a more nuanced understanding of tumor biology. This refined stratification sheds light on tumor heterogeneity and helps explain the variability in therapeutic responses observed among patients. Although BLBC is reported to exhibit high expression of immune checkpoint molecules such as CD274 and CD276, it is widely considered a cold tumor, as our findings show that most BLBC patients are non-responsive to immunotherapy [19]. This underscores the need for robust markers to identify BLBC and develop tailored therapeutic

strategies. Emerging evidence suggests that ALIX, through its regulation of PD-L1 and EGFR, may serve as a diagnostic marker and therapeutic target for BLBC [20]. Additionally, our findings showed that BLBC tumors overexpress VIM (vimentin), a marker associated with epithelial-to-mesenchymal transition (EMT). Given the established role of EMT in promoting tumor invasion and metastasis, targeting VIM may provide a novel avenue to reduce the metastatic potential of BLBC tumors.

The molecular signature defining our BLBC subpopulation provides crucial insights into the biological underpinnings of its aggressiveness. The marker genes we identified—including VIM, FSCN1, CRYAB, and RARRES1—are not merely correlative but are functionally implicated in core malignant processes. Specifically, VIM (vimentin) is recognized as a positive regulator of stemness in both the normal mammary gland and breast cancer, providing a developmental link to the heightened plasticity and self-renewal capacity observed in basal-like cells [21]. This stem-like property is further reinforced by FSCN1, which not only drives invasion and metastasis but also supports cancer cell self-renewal and drug resistance [22]. Simultaneously, CRYAB (α B-crystallin) confers a critical survival advantage, as its expression is linked to increased cell motility following chemotherapy in basal-like models, directly explaining the treatment-induced adaptation and metastatic propensity seen in this subtype [23]. The retention of RARRES1 expression, potentially regulated by subtype-specific mechanisms such as promoter hypomethylation and retinoic acid signaling, adds another layer of complexity, suggesting context-dependent tumor-suppressive or adaptive functions [24]. Collectively, this gene set delineates a BLBC phenotype characterized by intrinsic stemness, enhanced motility, and a formidable capacity to withstand therapeutic stress, thereby elucidating the molecular basis for its poor prognosis and resistance to conventional and immunotherapeutic regimens.

Critically, this highly malignant phenotype is intimately linked to a profoundly immunosuppressive TME. Our analysis revealed a significant enrichment of Tregs in high-PIS tumors, providing a crucial cellular mechanism for their immunosuppressive nature. This observation aligns with established literature wherein Treg recruitment is recognized as a hallmark of aggressive basal-like breast cancers and is correlated with poor prognosis [25,26]. The presence of Tregs within the TME is not passive; they actively orchestrate immune evasion by directly suppressing cytotoxic T-cell function and secreting inhibitory cytokines [27]. Importantly, our data situate this phenomenon within a broader adaptive immune resistance mechanism. The concurrent elevation of immune checkpoints in high-PIS tumors, suggests a synergistic immunosuppressive axis. This is supported by evidence that BLBC cells, upon sensing T-cell-derived IFN- γ , can upregulate PD-L1/PD-L2, which in turn promotes Treg differentiation and further dampens antitumor immunity [28,29]. Thus, the high-PIS phenotype appears to engender a self-reinforcing immunosuppressive circuit: an initial, perhaps abortive, immune response creates an inflammatory milieu that simultaneously recruits Tregs and upregulates checkpoint molecules, ultimately leading to T-cell exhaustion and dysfunction. This model explains the poor immunotherapy response observed in high-PIS patients and underscores the potential necessity of combining PD-1/PD-L1 blockade with Treg-targeting strategies to effectively reverse the profound immune suppression in this aggressive tumor subset.

Cycling tumor cells (CTCs) are considered among the earliest metastatic precursors, yet they often fail to represent whole-tumor characteristics due to spatial and temporal heterogeneity [30]. Pseudotime trajectory analysis revealed developmental connections between CTCs and BLBC, suggesting a shared evolutionary or functional origin. Prior studies have indicated that CTC clusters and polyclonal metastases in BLBC are driven by CD44+ cancer stem cells (CSCs), which are strongly associated with poor prognosis [31,32]. Furthermore, the co-occurrence of circulating tumor microemboli (CTMs) and CTCs has been observed, particularly in early-stage TNBC, implicating CTMs in the metastatic cascade [33]. These findings underscore the importance of

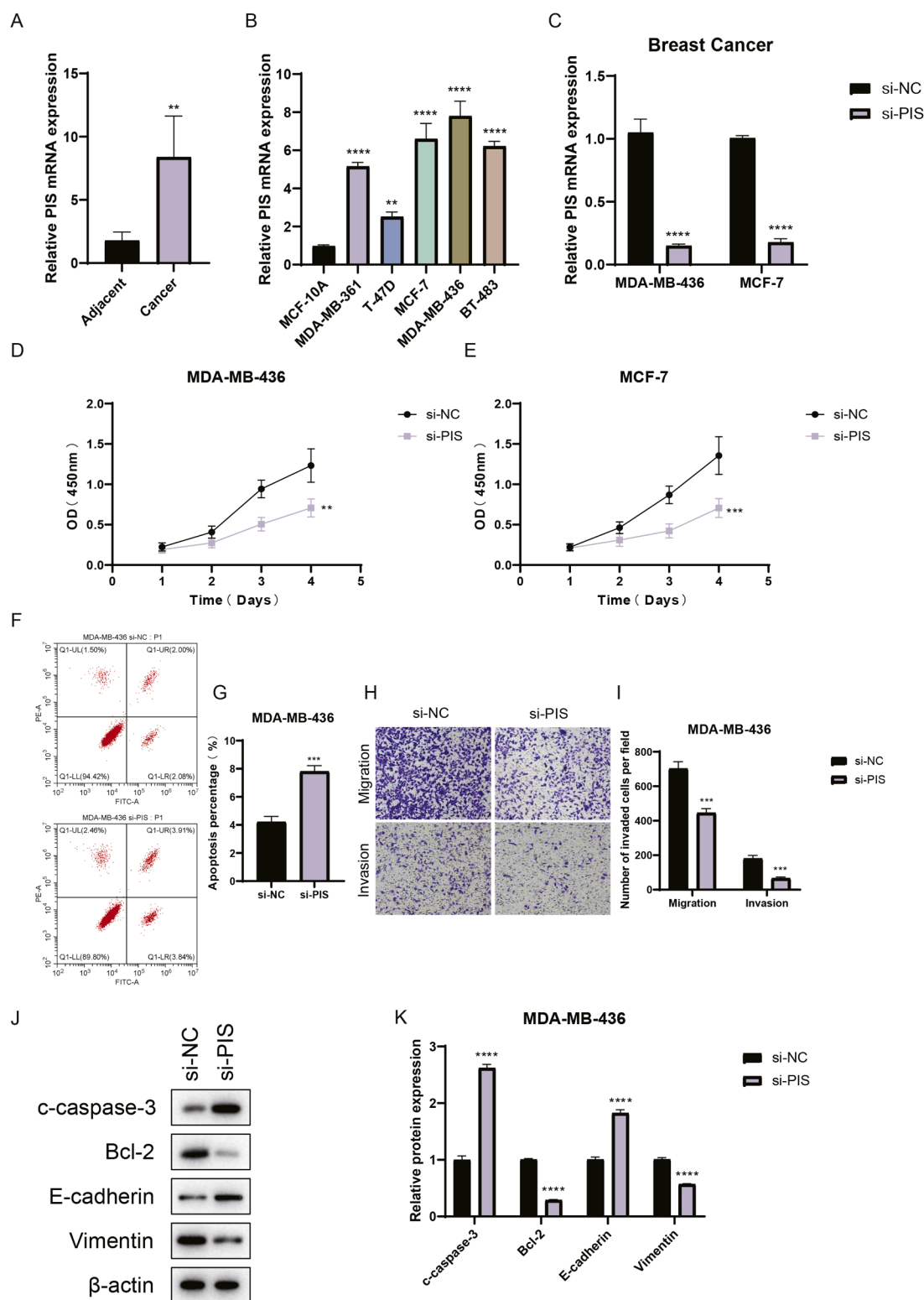


Fig. 11. Expression and functional analysis of PIS in breast cancer. (A) Relative mRNA expression of PIS in breast cancer tissues and matched adjacent normal tissues. (B) PIS mRNA expression levels in breast cancer cell lines (MDA-MB-361, T-47D, MCF7, MD-AMB-436, BT-483) related to MCF-10A. (C) qRT-PCR validation of PIS knockdown efficiency in MCF7 and MDA-MB-436 cells following siRNA transfection. (D–E) CCK-8 assays showing cell proliferation over 4 days in control and PIS knockdown groups (n = 5 per group). (F–G) Flow cytometry analysis of apoptosis in control and PIS-silenced MDA-MB-436 cells. (H–I) Transwell migration and invasion assays in MDA-MB-436 cells following PIS knockdown. (J–K) Western blot analysis of cleaved caspase-3, Bcl-2, E-cadherin, and Vimentin expression in control and PIS knockdown MDA-MB-436 cells.

characterizing CTCs and CTMs at a molecular level to understand their role in metastasis and develop targeted interventions.

Recent evidence challenges the traditional dichotomy between basal and luminal subtypes by demonstrating basal-like features in ER+ breast tumors [34]. Our spatial transcriptomics analysis revealed basal-like cells even within ER+ tumors, suggesting their pervasive presence across clinical subtypes. This phenomenon extends beyond ER+ disease, as evidenced by multi-omics profiling identifying a basal/mesenchymal-like (HER2-BM) subtype within HER2-positive breast cancer, which comprises over 20% of cases and exhibits poor response to standard anti-HER2 therapy [35]. The convergence of these findings across independent studies underscores a fundamental biological principle: the basal-like phenotype represents a potent, trans-subtype oncogenic driver that can emerge across the spectrum of clinical classifications, fundamentally redefining our understanding of breast cancer heterogeneity. This discovery aligns with reports of basal markers such as CK5/6 and EGFR being expressed in subsets of ER+ tumors, further complicating subtype classification [36]. Studies have identified nuclear CSPP1 isoforms in basal-like carcinomas with luminal traits, reinforcing the complexity of tumor subtype interplay [37]. Critically, the presence of these basal-like elements, whether as a minor subpopulation in ER+ tumors or defining an entire HER2+ molecular subtype, is not merely a molecular curiosity but a key determinant of clinical behavior, underlying aggressive tumor progression and intrinsic resistance to conventional therapies. These findings suggest that basal-like cells within ER+ tumors may contribute to aggressive behavior and impact treatment outcomes, highlighting the need for refined classification and therapeutic strategies. Our development of the PIS directly addresses this need by providing a quantitative framework to capture this critical biological axis, thereby paving the way for treatment strategies that move beyond conventional subtypes to target the underlying basal-like biology, irrespective of a tumor's ER or HER2 status.

Notwithstanding the insights gained, certain limitations of our study warrant consideration. The transcriptional focus of our analysis, while revealing, underscores the need for complementary proteomic and metabolomic profiling to fully elucidate the functional mechanisms underlying the aggressive phenotype of basal-like subpopulations. In addition, the deconvolution of immune cell infiltration and the PIS model, though rigorously validated across multiple cohorts, are inherently constrained by the computational inference of cellular composition from bulk transcriptomic data. Future investigations would benefit from incorporating multi-omics approaches in expanded clinical cohorts to further substantiate the prognostic utility of PIS, as well as functional studies to dissect the causal contributions of basal-like cells to immune suppression and treatment resistance.

In conclusion, this study significantly advances our understanding of breast cancer heterogeneity and its implications for immunotherapy. By identifying BLBC as a major contributor to immune evasion and therapy resistance and through the development of the PIS model, we provide actionable insights for precision treatment. Future research should prioritize integrating multi-omics and spatial transcriptomic data to further unravel tumor heterogeneity and target high-risk tumor subpopulations with innovative therapeutic strategies.

Data availability statement

The datasets used in this study can be found in online repositories.

Funding

This work was supported by the Zhejiang Provincial Key Innovation Team for Community-Acquired Severe Pneumonia (CXTD202501057) and the Natural Science Foundation of Shandong Province (ZR2022QH049).

CRediT authorship contribution statement

Zhenxiong Zhao: Writing – original draft, Visualization, Validation. **Zhencang Zheng:** Writing – original draft, Project administration, Investigation, Formal analysis, Data curation. **Shenglu Jiang:** Writing – original draft, Validation, Supervision, Software, Conceptualization. **Lingling Zhang:** Writing – original draft, Visualization. **Xiufeng Tang:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no conflicts of interest to report regarding the present study.

References

- [1] Giannakeas, V., Lim, D. W. & Narod, S. A. Bilateral mastectomy and breast cancer mortality. (2024).
- [2] Yu, J. Challenges and opportunities in metastatic breast cancer treatments: nano-drug combinations delivered preferentially to metastatic cells may enhance therapeutic response. (2022).
- [3] R. Feng, et al., Cancer situation in China: what does the China cancer map indicate from the first national death survey to the latest cancer registration? *Cancer Commun.* 43 (2023) 75–86.
- [4] Y. Li, et al., Recent advances in therapeutic strategies for triple-negative breast cancer, *J. Hematol. Oncol.* 15 (2022) 121.
- [5] L. Kraehenbuehl, C.-H. Weng, S. Eghbali, J.D. Wolchok, T. Merghoub, Enhancing immunotherapy in cancer by targeting emerging immunomodulatory pathways, *Nat. Rev. Clin. Oncol.* 19 (2022) 37–50.
- [6] S. Zhu, et al., Recent advances in targeted strategies for triple-negative breast cancer, *J. Hematol. Oncol.* 16 (2023) 100.
- [7] E. Agostinotto, et al., Progress and pitfalls in the use of immunotherapy for patients with triple negative breast cancer, *Expert. Opin. Investig. Drugs* 31 (2022) 567–591.
- [8] H. Yan, et al., Clinical evidence for efficacy of pembrolizumab in MSI-H and TMB-H advanced solid tumor: results from three cancer centers in China, *Cancer Immunol. Immunother.* 73 (2024) 74.
- [9] M. Rakae, et al., Association of Machine Learning-Based Assessment of Tumor-Infiltrating Lymphocytes on Standard Histologic Images With Outcomes of Immunotherapy in Patients With NSCLC, *JAMA Oncol.* 9 (2023) 51.
- [10] Ding, S., Chen, X. & Shen, K. Single-cell RNA sequencing in breast cancer: Understanding tumor heterogeneity and paving roads to individualized therapy.
- [11] V. Kast, et al., A tumor microenvironment model of pancreatic cancer to elucidate responses toward immunotherapy, *Adv. Healthc. Mater.* 12 (2023) 2201907.
- [12] T. Ge, et al., Crosstalk between metabolic reprogramming and epigenetics in cancer: updates on mechanisms and therapeutic opportunities, *Cancer Commun.* 42 (2022) 1049–1082.
- [13] T. Yasuda, Y.A. Wang, Gastric cancer immunosuppressive microenvironment heterogeneity: implications for therapy development, *Trends. Cancer* 10 (2024) 627–642.
- [14] S.-Q. Liu, et al., Single-cell and spatially resolved analysis uncovers cell heterogeneity of breast cancer, *J. Hematol. Oncol.* 15 (2022) 19.
- [15] Anagnostou, V. The status of tumor mutational burden and immunotherapy. 3, (2022).
- [16] S. Suma, et al., Tumor heterogeneity and immune-evasive T follicular cell lymphoma phenotypes at single-cell resolution, *Leukemia* 38 (2024) 340–350.
- [17] G. Gargiulo, M. Serresi, J.-C. Marine, Cell states in cancer: drivers, passengers, and trailers, *Cancer Discov.* 14 (2024) 610–614.
- [18] M.D. Vesely, T. Zhang, L. Chen, Resistance mechanisms to anti-PD cancer immunotherapy, *Annu. Rev. Immunol.* 40 (2022) 45–74.
- [19] J. Zhao, J. Huang, Breast cancer immunology and immunotherapy: targeting the programmed cell death protein-1/programmed cell death protein ligand-1, *Chin. Med. J. (Engl)* 133 (2020) 853–862.
- [20] J. Monypenny, et al., ALIX regulates tumor-mediated immunosuppression by controlling EGFR activity and PD-L1 presentation, *Cell Rep.* 24 (2018) 630–641.
- [21] E. Peuhu, R. Virtakoivu, A. Mai, A. Wärr, J. Ivaska, Epithelial vimentin plays a functional role in mammary gland development, *Development* 144 (2017) 4103–4113.
- [22] D. Chang, X.-X. Liu, R. Liu, J.-W. Sun, The role and regulatory mechanism of FSCN1 in breast tumorigenesis and progression, *Yi. Chuan.* 45 (2023) 115–127.
- [23] L. Yang, et al., Fluorouracil exacerbates alpha-crystallin B chain-mediated cell migration in triple-negative breast cancer cell lines, *Sci. Rep.* 13 (2023) 4010.
- [24] K.M. Coyle, et al., Breast cancer subtype dictates DNA methylation and ALDH1A3-mediated expression of tumor suppressor RARRES1, *Oncotarget.* 7 (2016) 44096–44112.
- [25] S.J. Kim, S. Garcia-Recio, C.J. Creighton, C.M. Perou, J.M. Rosen, Alterations in Wnt- and/or STAT3 signaling pathways and the immune microenvironment during metastatic progression, *Oncogene* 38 (2019) 5942–5958.
- [26] M. Yan, et al., Recruitment of regulatory T cells is correlated with hypoxia-induced CXCR4 expression, and is associated with poor prognosis in basal-like breast cancers, *Breast. Cancer Res.* 13 (2011) R47.

- [27] S. Bhattacharya, G. Paraskar, M. Jha, G.L. Gupta, B.G. Prajapati, Deciphering regulatory T-cell dynamics in cancer immunotherapy: mechanisms, implications, and therapeutic innovations, *ACS. Pharmacol. Transl. Sci.* 7 (2024) 2215–2236.
- [28] P. Karasar, G. Esendagli, T helper responses are maintained by basal-like breast cancer cells and confer to immune modulation via upregulation of PD-1 ligands, *Breast. Cancer Res. Treat.* 145 (2014) 605–614.
- [29] Z. Li, et al., PD-L1 expression is associated with tumor FOXP3⁺ regulatory T-cell infiltration of breast cancer and poor prognosis of patient, *J. Cancer* 7 (2016) 784–793.
- [30] D. Lin, et al., Circulating tumor cells: biology and clinical significance, *Sig. Transduct. Target Ther.* 6 (2021) 404.
- [31] Liu, X. et al. Homophilic CD44 interactions mediate tumor cell aggregation and polyclonal metastasis in patient-derived breast cancer models. (2020).
- [32] J.M. Balko, et al., Activation of MAPK pathways due to DUSP4 loss promotes cancer stem cell-like phenotypes in basal-like breast cancer, *Cancer Res.* 73 (2013) 6346–6358.
- [33] Reduzzi, C. et al. Circulating tumor cell clusters are frequently detected in women with early-stage breast cancer. (2021).
- [34] Fan, M. Triggering a switch from basal- to luminal-like breast cancer subtype by the small-molecule diptoinonesin G via induction of GABARAPL1. (2020).
- [35] Y.-W. Li, et al., Molecular characterization and classification of HER2-positive breast cancer inform tailored therapeutic strategies, *Cancer Res.* 84 (2024) 3669–3683.
- [36] C. Choccalingam, L. Rao, S. Rao, Clinico-pathological characteristics of triple negative and non triple negative high grade breast carcinomas with and without basal marker (CK5/6 and EGFR) expression at a rural tertiary hospital in India, *Breast Cancer (Auckl)* 6 (2012) S8611. BCBCR.
- [37] J. Sternemalm, et al., Nuclear CSPP1 expression defined subtypes of basal-like breast cancer, *Br. J. Cancer* 111 (2014) 326–338.